Cloning, Expression, and Functional Analysis of the Destabilised L135P Missense Variant of Catechol-O-Methyltransferase (COMT) in *Escherichia coli*

1. Objective

This project aims to investigate the functional consequences of the rare missense variant L135P on the human soluble catechol-O-methyltransferase (S-COMT) enzyme. The study will focus on the L135P missense variant, which has been associated with severe structural destabilisation and proteasomal degradation. This particular variant was selected because while its impact on protein stability is known, its intrinsic catalytic properties have not been characterised, representing a key knowledge gap (Larsen *et al.*, 2023). This project aims to determine whether the L135P mutation alters the catalytic efficiency of S-COMT independently of its effect on protein stability. A direct enzymatic assay will be used to compare the Michaelis–Menten kinetic parameters (K_m and V_{max}) of recombinant wild-type and L135P S-COMT, contributing to a better understanding of how rare genetic variation can impair COMT function.

2. Background and Rationale

Catechol-O-methyltransferase (COMT) is a ubiquitous mammalian enzyme (EC 2.1.1.6) that plays a crucial role in the metabolic inactivation of catechol-containing compounds. COMT acts in the body by catalysing the transfer of a methyl group from the donor S-adenosyl-L-methionine (SAM) to a hydroxyl group of its substrates, a reaction that requires a divalent magnesium ion (Mg^{2+}) as a cofactor (Vidgren *et al.*, 1994). Its role in neurotransmitter metabolism is particularly important, as its substrates include the catecholamine neurotransmitters dopamine, norepinephrine, and epinephrine (Männistö and Kaakkola, 1999).

Dysregulation of COMT activity has been implicated in various physiological and pathological states. COMT is particularly important in the prefrontal cortex, where it represents the primary mechanism for dopamine clearance due to the low density of dopamine transporters in this region (Chen *et al.*, 2004). Consequently, altered COMT function can affect cognitive processes and has been explored as a factor in the pathophysiology of several neuropsychiatric conditions, such as schizophrenia, bipolar disorder, and anxiety disorders (Witte and Flöel, 2012). Clinically, COMT inhibitors are used as an adjunct therapy in the treatment of Parkinson’s disease,

where they prevent the peripheral degradation of the drug L-DOPA, thereby increasing its bioavailability to the brain (Männistö and Kaakkola, 1999; Akhtar *et al.*, 2020).

COMT polymorphisms, particularly missense mutations, have been associated with significant variations in enzyme activity. Missense mutations are single nucleotide changes that result in a different amino acid being incorporated into the protein, which can affect protein structure and function. Certain missense variants result in reduced COMT activity by decreasing conformational stability and promoting protein turnover. The most studied example is the Val108Met polymorphism (rs4680) (corresponding to Val158Met in the membrane-bound isoform), which results in a three- to four-fold reduction in enzyme activity due to decreased thermostability (Lotta *et al.*, 1995). Structural studies have shown that the Val-to-Met substitution does not alter the active site geometry directly but reduces the overall conformational stability of the protein (Rutherford *et al.*, 2008). However, the full spectrum of functional genetic variation extends well beyond this common polymorphism, with numerous other non-synonymous variants having been identified through gene resequencing (Shield *et al.*, 2004).

A particularly relevant variant is L135P, a rare missense mutation in which leucine at position 135 is replaced by proline. This variant leads to profound structural destabilisation, causing the protein to be targeted for proteasomal degradation in human cells (Larsen *et al.*, 2023). In an *E. coli* expression system, which lacks a proteasome, this instability would likely result in reduced soluble protein yield, increased aggregation into inclusion bodies, or degradation by bacterial proteases. Understanding the biochemical properties of this variant is therefore of significant interest. To investigate the impact of the L135P mutation on intrinsic enzyme function, recombinant COMT proteins will be produced. Recombinant protein expression is required to generate the milligram quantities of purified protein necessary for detailed *in vitro* kinetic characterisation, bypassing the confounding variable of cellular protein degradation.

The pET28a vector from the pET-series will be utilised in this study because it is a high-level expression vector specifically designed for use in *E. coli*. This vector features a strong, IPTG-inducible T7 promoter system, which allows for tightly controlled, high-level protein production (Shilling *et al.*, 2020). Furthermore, pET28a

offers an N-terminal hexahistidine (6xHis) affinity tag, which enables a straightforward, single-step purification of the recombinant protein by immobilised metal affinity chromatography (IMAC) (Spriestersbach *et al.*, 2015).

3. Gene Selection and Cloning Strategy

The soluble catechol-O-methyltransferase (S-COMT) isoform will be used in this study because it is the predominant form in most tissues and the isoform used in the majority of published structural and kinetic studies (Lotta *et al.*, 1995). The gene sequence is sourced from the National Center for Biotechnology Information (NCBI, Gene ID: 1312), using the MANE Select transcript NM_000754.4 as the reference. While this transcript's primary CDS encodes the longer membrane-bound (MB-COMT) isoform, the shorter soluble (S-COMT) isoform is produced via an alternative downstream translation initiation site. This project will therefore clone the 221-amino acid S-COMT open reading frame to avoid the N-terminal membrane-anchoring region. A commercially synthesised DNA sequence of the S-COMT ORF corresponding to the native human coding sequence (RefSeq NM_000754.4; 221 aa) will be purchased from a commercial supplier (GenScript).

The pET28a vector will be purchased from Merck (formerly Novagen). The cloning strategy will be based on a standard restriction–ligation approach. The S-COMT DNA fragment and pET28a vector will be digested with NdeI and XhoI restriction enzymes (New England Biolabs). This design facilitates directional cloning and ensures that the gene is in-frame with the vector-encoded N-terminal 6xHis-tag. The forward primer incorporates the NdeI site (CATATG), which contains the start codon, along with a 6-nt protective clamp (5′-GCGCGC**CATATG**GGTGACACCAAGGAGCAGCG-3′), while the reverse primer includes the XhoI site (CTCGAG) immediately following the stop codon (5′-GCGCGC**CTCGAG**TCAGGGCCCTGCTTCGCTGC-3′). Cloning primers were designed by selecting gene-specific flanking regions and appending restriction sites and 5′ clamps, then validated for T_m using NEB T_m calculator prior to ordering from Geneviz.

After PCR amplification from the synthesised template, the products will be purified using a commercial PCR clean-up kit following the manufacturer's guidelines. The S-COMT DNA fragment and pET28a vector will then be digested with NdeI and XhoI according to the manufacturer's recommendations. This will then be ligated using T4

DNA ligase, following the standard ligation protocol. The ligation product will be transformed into chemically competent *E. coli* DH5α cells for plasmid propagation. Purification of successful clones will be carried out using a plasmid miniprep kit. This will further be confirmed by Sanger sequencing using T7 promoter and T7 terminator primers to verify the entire insert sequence.

4. Missense Variant Selection and Mutagenesis Design

To introduce the L135P mutation into the wild-type S-COMT clone, a PCR-based site-directed mutagenesis (SDM) strategy will be employed, following the principles of the QuikChange protocol (Agilent Technologies, 2005). This approach enables the precise introduction of the c.404T>C nucleotide change (based on the native S-COMT CDS) that converts the leucine codon (CTG) to a proline codon (CCG) at position 135. Oligonucleotide primers will be designed to target the codon encoding Leu135, with the mismatched base centred (target Tm ~78–80°C), designed using NEBaseChanger (NEB) and synthesised by Geneviz (forward: 5′-ATGATGTGGACACACCGGACATGGTCTTCCT-3′, reverse:5′-AGGAAGACCATGTCCGGTGTGTCCACATCAT-3′). The introduction of proline, a known disruptor of secondary structure, is predicted to severely impact the protein fold (Larsen *et al.*, 2023).

Following high-fidelity PCR amplification of the entire plasmid with mutagenic primers, the reaction mixture will be treated with the DpnI restriction enzyme, which selectively digests the methylated parental (wild-type) DNA, leaving only the unmethylated, newly synthesised mutant plasmid. The resulting nicked plasmid DNA will be transformed into *E. coli*, where the nicks are repaired. The mutation will be verified by complete sequencing of the S-COMT coding region to ensure only the intended change has been made.

5. Expression Protocol in E. coli

The wild-type and mutant COMT constructs cloned into pET28a will be expressed in the *E. coli* host strain BL21(DE3). This strain is used for recombinant COMT expression because it contains a chromosomal copy of the T7 RNA polymerase gene under the control of the IPTG-inducible *lacUV5* promoter, which is required for transcription from the pET vector's T7 promoter (Shilling *et al.*, 2020). Furthermore, it

is deficient in Lon and OmpT proteases, which helps to improve the yield of intact recombinant proteins (Francis and Page, 2010).

Verified plasmid constructs will be transformed into chemically competent BL21(DE3) cells by heat shock at 42°C for 45 seconds, followed by recovery in SOC medium at 37°C for 1 hour. Transformants will be selected on LB agar plates supplemented with 50 µg/mL kanamycin, the selectable marker encoded on pET28a. A single colony will be used to inoculate a starter culture, which will then be used to inoculate a 1 L culture. The culture will be grown at 37°C with shaking to an OD_{600} of 0.6–0.8. The role of IPTG is to induce expression of T7 RNA polymerase; upon reaching the desired cell density, expression will be induced by adding IPTG to a final concentration of 0.5 mM. Following induction, cells will be incubated for 16–18 hours at a reduced temperature of 18°C to promote correct protein folding (Rosano and Ceccarelli, 2014). Cells will be harvested by centrifugation (5,000 × g, 15 min, 4°C), and pellets stored at −80°C until purification. Pre- and post-induction samples will be retained for SDS-PAGE to verify expression and solubility.

6. Protein Purification Protocol

To purify the recombinant S-COMT, including the mutant, an affinity chromatography strategy will be used. The hexahistidine tag encoded by pET28a has a high affinity for immobilised Ni^{2+} ions coordinated on nitrilotriacetic acid (NTA) resin, enabling selective capture of the tagged protein from crude lysate (Spriestersbach *et al.*, 2015). Purification will therefore be performed by Ni-NTA IMAC under native conditions. All steps will be performed at 4°C. Cell pellets will be resuspended in ~30 mL lysis buffer (50 mM sodium phosphate pH 8.0; 300 mM NaCl; 10 mM imidazole) supplemented with EDTA-free protease inhibitor, lysozyme (1 mg/mL) and DNase I (10 µg/mL). After 30 min on ice, cells will be disrupted by sonication. The lysate will be clarified by centrifugation (20,000 × g, 30 min, 4°C) and the soluble fraction loaded onto a pre-equilibrated Ni-NTA column. The column will be washed with ~20 column volumes of wash buffer (20 mM imidazole) to remove non-specific proteins, then His-tagged S-COMT will be eluted with 250 mM imidazole. Soluble and insoluble fractions will be analysed by SDS-PAGE to quantify solubility differences between WT and L135P.

Protein concentration will be determined using the Bradford assay with BSA standards. Pooled elution fractions will be dialysed overnight at 4°C into storage buffer (50 mM Tris-HCl pH 7.4; 150 mM NaCl; 10% glycerol) to remove imidazole. SDS-PAGE will be used to confirm the purity and integrity of the recombinant COMT expression and visualised using Coomassie staining, with a distinct band at approximately 26–27 kDa confirming successful COMT expression. For further validation, Western blotting using an anti-His antibody will be performed following transfer to a nitrocellulose membrane. Purified proteins will be separated and stored at −80°C.

7. Functional Analysis of Wild-Type and Mutant COMT

To evaluate the enzymatic activity of wild-type (WT) and L135P mutant COMT, an HPLC-based assay will be used, following the method described by Reenilä, Tuomainen and Männistö (1995). This assay measures COMT-catalysed transfer of a methyl group from SAM to a catechol substrate by quantifying methylated products. Given that the L135P mutation has been reported to cause severe structural destabilisation (Larsen *et al.*, 2023), this experiment will determine whether the mutation also alters intrinsic catalytic efficiency (K_m, V_{max}, k_{cat}/K_m) independent of protein stability. It is hypothesised that L135P will show impaired catalytic performance (increased K_m and/or decreased V_{max}) relative to WT.

The reaction will be conducted in 100 mM potassium phosphate buffer (pH 7.4) containing 5 mM $MgCl_2$ at 37°C. COMT activity will be quantified by measuring the formation of vanillic and isovanillic acid from the substrate 3,4-dihydroxybenzoic acid (DHBAc) in the presence of SAM, with products quantified by HPLC with electrochemical detection using external standards. To estimate K_m for DHBAc, the substrate will be titrated (1–1000 µM) while SAM is held at a saturating concentration (500 µM). Reactions will be initiated by addition of purified enzyme (1–5 µg), incubated within the linear range of product formation (~20 min), and quenched on ice with perchloric acid. Negative controls will include no-enzyme, no-SAM, and heat-inactivated enzyme reactions to confirm that activity is enzymatic, SAM-dependent, and requires correctly folded protein.

The kinetic properties of the purified COMT will be assessed by calculating initial velocities across the DHBAc titration and fitting data to the Michaelis–Menten

equation using non-linear regression to obtain K_m and V_{max}. k_{cat} will be calculated from V_{max} and the molar enzyme concentration. All reactions will be run in triplicate, and parameters compared across ≥3 independent protein preparations. Statistical significance will be assessed using an unpaired t-test.

8. Expected Results and Potential Challenges

Given that substitutions at sites like Leu135 can be highly disruptive, the variant is likely to exhibit significantly reduced soluble expression and impaired catalytic activity. As a result, the mutant enzyme may exhibit reduced soluble yield and increased aggregation into inclusion bodies. In enzymatic assays, the L135P mutant is anticipated to show either an increased K_m, a decreased V_{max}, or both, leading to a lower catalytic efficiency (k_{cat}/K_m) than WT. These changes could lead to an overall impairment consistent with a loss-of-function phenotype.

Two principal outcome scenarios can be distinguished. If soluble L135P protein is recovered at near-normal yield but kinetic parameters are impaired, this would indicate that the mutation primarily disrupts the active-site environment, representing intrinsic catalytic impairment. Conversely, if yield is severely reduced but the purified L135P displays near-normal specific activity, the dominant mechanism would be protein instability rather than catalytic deficiency. A combination of both would support a dual loss-of-function mechanism (Larsen *et al.*, 2023).

The main anticipated challenge is poor solubility of the L135P variant. Mitigations include lowering induction strength (IPTG 0.1–0.25 mM), reducing temperature, and co-expressing molecular chaperones. If the protein remains insoluble, denaturing solubilisation (8 M urea) followed by on-column refolding can be attempted. A second risk is complete loss of measurable activity. To address this, protein identity will be re-checked by SDS-PAGE and anti-His western blotting, assay components will be validated with WT enzyme, and the linear range will be re-established. If inactivity persists, circular dichroism (CD) spectroscopy can assess whether the purified L135P material is grossly misfolded compared to wild-type.

9. References

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